STEP 2 — Dependencies Install Karna

Time: ~10 minutes · Internet required · Is step ka output: saare Python libraries install

> **Sabse pehle samajh lo: virtual environment (venv)** ek alag “dabba” hai jisme sirf is project ki libraries install hoti hain. Isse aapke computer ki baaki Python settings kharab nahi hoti. **Har baar project kholne par is dabbe ko activate karna padta hai.**

2.1 Virtual environment banana

VS Code ka terminal kholiye (CareFlow folder me) aur ye chala dijiye:

Windows par

```
python -m venv .venv
```

Mac / Linux par

```
python3 -m venv .venv
```

Kuch seconds lagenge. Ab left panel me ek naya hidden folder `.venv` dikhega — yeh aapka “dabba” hai. Ise delete mat kijiye.

2.2 Virtual environment activate karna

Windows — PowerShell

```
.\.venv\Scripts\Activate.ps1
```

Agar error aaye: “running scripts is disabled on this system”, to pehle ye chala dijiye:

```
Set-ExecutionPolicy -ExecutionPolicy RemoteSigned -Scope CurrentUser
```

Windows — Command Prompt (CMD)

```
.venv\Scripts\activate.bat
```

Mac / Linux

```
source .venv/bin/activate
```

Activate hua ya nahi — pehchan

Terminal prompt ke aage `(.venv)` likha dikhne lagega:

```
(.venv) C:\Projects\CareFlow>
```

Yeh `(.venv)` dikhna zaroori hai. Nah dikhe to aap system Python me install kar rahe ho aur baad me “module not found” errors aayenge.

2.3 Core libraries install karna

```
pip install --upgrade pip
pip install -r requirements.txt
```

Yeh command `requirements.txt` me likhi libraries install karegi. Isme kya-kya hai:

Library	Kaam kya karti hai
`pandas`	Data load, clean aur analyse karna (event log ka dil)
`numpy`	Fast maths calculation
`pyyaml`	`config/config.yaml` padhna
`matplotlib` + `seaborn`	14 static charts (PNG) banana
`networkx`	Process model / Directly-Follows Graph draw karna
`scikit-learn`	Long-stay prediction model train karna
`plotly`	Dashboard ke interactive charts
`streamlit`	Interactive web dashboard
`openpyxl`	Excel export support
`reportlab`	PDF export support
`joblib`	Trained model ko file me save karna

Install hone me 1–3 minute lagte hain. Sab ke last me aisa dikhna chahiye:

```
Successfully installed altair-5.x joblib-1.x matplotlib-3.x networkx-3.x
pandas-2.x plotly-5.x scikit-learn-1.x seaborn-0.x streamlit-1.x ...
```

2.4 Install check karna

```
python -c "import pandas, numpy, sklearn, networkx, seaborn, plotly, streamlit, joblib,
yaml, matplotlib; print('ALL CORE LIBRARIES OK')"
```

Output aana chahiye:

```
ALL CORE LIBRARIES OK
```

Agar koi library missing dikhe, to sirf wahi install kijiye, jaise:
`pip install seaborn` ya `pip install joblib plotly`

2.5 OPTIONAL: pm4py install karna (recommended)

`pm4py` duniya ki sabse badi open-source **process mining library** hai. Isse aapko official algorithms milte hain: **Inductive Miner**, **Heuristics Miner**, **Alpha Miner** aur **Petri net** export.

> **Zaroori baat:** CareFlow **pm4py ke bina bhi poora chalta hai**. `src/discover_models.py` ke andar ek pure pandas + networkx discovery engine already likha hua hai. pm4py install karenge to result me extra Petri-net model bhi banega — report me "official algorithm" likhne ka bonus milta hai.

```
pip install -r requirements-optional.txt
```

Ya directly:

```
pip install pm4py graphviz
```

Graphviz binaries (petri net image ke liye zaroori)

OS	Kya karna hai
Windows	`https://graphviz.org/download/` se **64-bit EXE installer** download karke install kijiye.
Windows (PATH)	Install ke baad: Start → "Environment Variables" → `Path` → Edit → New → `C:\Program Files\Graphviz\bin` → OK. Terminal band karke dobara kholiye.
Mac	`brew install graphviz`
Ubuntu	`sudo apt-get install -y graphviz`

pm4py aur graphviz check karna

```
python -c "import pm4py; print('pm4py', pm4py.__version__)"
dot -V
```

```
pm4py 2.7.x
dot - graphviz version 11.0.0
```

> Agar `import pm4py` fail ho jaye to **koi tension nahi** — `python src/discover_models.py` chalega aur log me likhega `pm4py not installed - using the built-in pandas/networkx engine`. Project pura complete rahega.

2.6 Streamlit check karna

```
streamlit hello
```

Browser khulega `http://localhost:8501` par, ek demo page dikhega. Terminal me `Ctrl + C` daba kar band kijiye. Streamlit sahi hai.

2.7 Requirements file ki poori list

`requirements.txt` (core — inhi se project chalta hai)

```
pandas>=2.0.0
numpy>=1.24.0
pyyaml>=6.0
matplotlib>=3.7.0
networkx>=3.1
scikit-learn>=1.3.0
plotly>=5.18.0
streamlit>=1.32.0
openpyxl>=3.1.0
reportlab>=4.0.0
```

`requirements-optional.txt` (optional — official process-mining algorithms)

```
pm4py>=2.7.11
graphviz>=0.20.1
```

2.8 Troubleshooting — common errors

Error	Matlab	Solution
`pip is not recognized`	venv activate nahi hai	Step 2.2 dobara kijiye, `(.venv)` dikhna chahiye
`No module named pandas`	galat Python use ho raha hai	`python -m pip install -r requirements.txt`
`running scripts is disabled`	PowerShell policy	`Set-ExecutionPolicy RemoteSigned -Scope CurrentUser`
install bahut slow chal raha hai	network	`pip install --timeout 100 -r requirements.txt`
`ImportError: libGL.so.1` (Linux)	system library missing	`sudo apt-get install -y libgl1 libglib2.0-0`
Graphviz `dot` not found	PATH set nahi	Step 2.5 ka PATH wala part dobara kijiye
Kuch bhi samajh nahi aa raha	—	venv delete karke (folder `.venv`) Step 2.1 se dobara shuru kijiye

Ho gaya! Libraries install hain, venv active hai. Ab Step 3 — data import.